

Brief Reports

Brief Reports are short papers which report on completed research which, while meeting the usual Physical Review standards of scientific quality, does not warrant a regular article. (Addenda to papers previously published in the Physical Review by the same authors are included in Brief Reports.) A Brief Report may be no longer than 3½ printed pages and must be accompanied by an abstract. The same publication schedule as for regular articles is followed, and page proofs are sent to authors.

Two continued-fractional treatments of multichannel scattering

M. Znojil

*Institute of Nuclear Physics, Czechoslovak Academy of Sciences,
250 68 Řež, Czechoslovakia
(Received 30 January 1984)*

A recent paper of Horáček and Sasakawa [Phys. Rev. A **28**, 2151 (1983)] is discussed. Firstly, we show that their formulation of the theory for multichannel scattering should be corrected. Then, after a slight modification, their continued-fractional one-channel formalism remains applicable to the multichannel case as well. Secondly, an alternative multichannel method is suggested based on a similar idea. It represents the solution of the corresponding coupled integral equations in terms of the matrix continued fractions.

In the various contexts including the atomic physics, most of the realistic scattering problems lead sooner or later to their coupled-channel formulation. Recently, the corresponding Lippmann-Schwinger integral equation

$$F_i = \phi_{i_0} \delta_{ii_0} + \sum_{j=1}^N G_i U_{ij} F_j, \quad i, i_0 = 1, 2, \dots, N, \quad (1)$$

$$F_i = F_i(E, r), \quad r \in (0, \infty),$$

has been shown to be solvable by the method of continued fractions (MCF, Ref. 1) for $N=1$. Our present intention is twofold: (1) to clarify certain inconsistency in the original $N > 1$ generalization of the method¹ (Sec. I); (2) to suggest an alternative approach based on the method of matrix continued fractions (Ref. 2, cf. Sec. II).

Both these methodical alternatives will prove to be related to our interpretation of i_0 in (1). This index characterizes the incident channel and may be considered either fixed (Sec. I, MCF) or variable ($i_0 = 1, 2, \dots, N$, Sec. II). In the latter case, our matrix form of MCF seems to be much more adequate since it uses only one recurrently defined Lanczos-type basis $u_{(i)}$ for all the incident channels i_0 .

I. CORRIGENDUM: MCF IN THE MULTICHANNEL CASE

The method of Horáček and Sasakawa¹ is an iterative prescription designed for the scattering equations of the type

$$|\varphi\rangle = |u\rangle + GU|\varphi\rangle. \quad (2)$$

Its algorithmic formulation is simple and the preliminary tests exhibit its high efficiency.¹ Their excellent convergence properties seem to be suitable for applications in the atomic physics.

Unfortunately, the authors treat merely the $N=1$ form of (1) in due detail, while the $N > 1$ formalism (Sec. III of Ref. 1) is rather confusing. Indeed, it contains a small but rather relevant inaccuracy in Eq. (35) where, besides the

function $\phi_i \delta_{ii_0}$, the inhomogeneous term should contain also the second component:

$$\sum_{j=1}^N [(1 - GU^{(1)})^{-1}]_{ij} G_j U_{j0}^{(1)} \phi_{i_0} \quad (3)$$

(the meaning of the symbols may be found in Ref. 1). In general, it vanishes for $N=1$ only. Vice versa, its presence makes the $N > 1$ continued-fractional construction invalid.

Fortunately, the whole problem is rather artificial and easily solvable. Indeed, we may complement the integration convention in (1), e.g.,

$$UF = \int_0^\infty dr r^2 U|r\rangle \langle r|F,$$

by the analogous summation convention concerning the channel indices, e.g.,

$$(UF)_i = \sum_{j=1}^N U_{ij} F_j.$$

Then, the dummy "double variable" $\zeta = (r, i)$ enters the $N > 1$ formalism in the same way as the radial coordinate does for $N=1$. Thus, modifying merely the notation in (1),

$$\begin{aligned} F_i &= \langle r, i | \varphi \rangle, \\ \phi_i \delta_{ii_0} &= \langle r, i | u \rangle (= 0 \text{ for } i \neq i_0), \\ U_{ij} &= \langle r, i | U | r', j \rangle, \\ G_i &= \langle r, i | G | r', j \rangle \delta_{ij}, \end{aligned} \quad (4)$$

we may drop the variables ζ and get precisely Eq. (2). Hence, Sec. II of Ref. 1 becomes applicable to the multichannel scattering as well.

As long as the incident channel remains fixed, it is necessary and sufficient to put $\phi_i = 0$ for all $i \neq i_0$ in Sec. III of Ref. 1. We reobtain the valid argumentation again. Moreover, the puzzling and set aside "discussions about the optimal choice of these functions^{1"} become entirely redundant from the formal point of view.

II. ADDENDUM: FORMALISM WITH MATRIX CONTINUED FRACTIONS

From the physical point of view, the Horáček's and Sasakawa's set $\phi_i \neq 0$ of the N different incident waves is specified usually as an input whenever the incident channels vary. When we incorporate this into the notation (4),

$$F_i = \langle r, i | \phi^{[i_0]} \rangle, \quad (5)$$

$$\phi_i \delta_{i_0} = \langle r, i | u^{[i_0]} \rangle$$

(U and G do not depend on i_0), we may introduce a new, "genuinely multichannel" form of the "weakened" operator²

$$U^{(1)} = U - \sum_{i_0, j_0=1}^N U | u^{[i_0]} \rangle (D^{-1})^{[i_0, j_0]} \langle u^{[j_0]} | U, \quad (6)$$

$$D^{[i_0, j_0]} = \langle u^{[i_0]} | U | u^{[j_0]} \rangle.$$

This differs from Eq. (31) in Ref. 1 but we still have the relations

$$U^{(1)} | u^{[i_0]} \rangle = 0, \quad i_0 = 1, 2, \dots, N$$

$$| \phi^{[i_0]} \rangle = | u^{[i_0]} \rangle + \sum_{j_0=1}^N | \phi^{[j_0]} \rangle g_{(k+1)}^{[j_0, i_0]} = | u^{[i_0]} \rangle + \sum_{j_0=1}^N | u^{[j_0]} \rangle g_{(k+1)}^{[j_0, i_0]} + \sum_{j_0, j_0'=1}^N | u^{[j_0]} \rangle g_{(k+2)}^{[j_0, j_0']} g_{(k+1)}^{[j_0', i_0]} + \dots, \quad (10)$$

$$g_{(k+1)}^{[j_0, i_0]} = \sum_{i_0'=1}^N (Q_{(k)}^{-1})^{[j_0, i_0']} D_{(k)}^{[i_0', i_0]},$$

$$Q_{(k)}^{[i_0, j_0]} = D_{(k)}^{[i_0, j_0]} - \langle u^{[i_0]} | U^{(k)} | \phi_{(k+1)}^{[j_0]} \rangle, \quad i_0, j_0 = 1, 2, \dots, N, \quad k = 0, 1, 2, \dots$$

This is the core of our present matrix generalization of MCF.

In the spirit of Ref. 2, let us now simplify once more the notation. Our main idea is to reinterpret any N -plet $a^{[j_0]}$, $j_0 = 1, 2, \dots, N$ as a new N -dimensional ket symbol $a | j_0 \rangle$. Then, we may rewrite (5),

$$F_i = \langle r, i | \phi | i_0 \rangle, \quad (11)$$

$$\phi_i \delta_{i_0} = \langle r, i | u | i_0 \rangle.$$

Employing the abbreviations

$$\langle u_{(i-1)}^{[k_0]} | U^{(i-1)} | \phi_{(i)}^{[l_0]} \rangle = \{k_0 | K_{(i)} | l_0 \},$$

$$\langle u_{(i-1)}^{[k_0]} | U^{(i-1)} | u_{(i)}^{[l_0]} \rangle = \{k_0 | A_{(i)} | l_0 \}, \quad (12)$$

$$\langle u_{(i)}^{[k_0]} | U^{(i)} | u_{(i)}^{[l_0]} \rangle = \{k_0 | D_{(i)} | l_0 \},$$

we may drop also the incident-channel indices and derive the recurrences

$$K_{(i)} = A_{(i)} + D_{(i)} \frac{1}{D_{(i)} - K_{(i+1)}} D_{(i)}, \quad (13)$$

from (10). Thus, we may consider finally the $N \times N$ -dimensional scattering matrix with the elements

$$\langle u^{[k_0]} | U | \phi^{[l_0]} \rangle = \{k_0 | K | l_0 \} \quad (14)$$

and define it, in full analogy with MCF [Eq. (26) of Ref. 1]

containing Eq. (8) of Ref. 1 as a special $N = 1$ case.

Similarly, we may consider the higher iterations ($U = U^{(0)}$, $| u_{(0)}^{[i_0]} \rangle = | u_{(0)}^{[i_0]} \rangle$) and postulate the $N > 1$ generalization of Eq. (21b) of Ref. 1,

$$| u_{(M+1)}^{[i_0]} \rangle = G U^{(M)} | u_{(M)}^{[i_0]} \rangle,$$

$$U^{(M+1)} = U^{(M)} - U^{(M)} \sum_{i_0, j_0=1}^N | u_{(M)}^{[i_0]} \rangle (D_{(M)}^{-1})^{[i_0, j_0]} \langle u_{(M)}^{[j_0]} | U^{(M)}, \quad (7)$$

$$D_{(M)}^{[i_0, j_0]} = \langle u_{(M)}^{[i_0]} | U^{(M)} | u_{(M)}^{[j_0]} \rangle, \quad M = 0, 1, 2, \dots$$

Again, we may expect that the operators $U^{(M)}$, $M \geq 1$ are "small" since they are zero in the larger and larger space,

$$U^{(M)} | u_{(0)}^{[i_0]} \rangle = 0, \quad (8)$$

$$i_0 = 1, 2, \dots, N, \quad i = 0, 1, \dots, M-1.$$

Using the definition analogous to Eq. (21c) of Ref. 1,

$$| \phi_{(M)}^{[i_0]} \rangle = (1 - G U^{(M)})^{-1} | u_{(M)}^{[i_0]} \rangle \quad (9)$$

(notice that $\phi_{(0)}^{[i_0]} = \phi^{[i_0]}$ is the desired solution), we may² derive the recurrent $N > 1$ counterparts of the iterative prescriptions (12) or (21a) of Ref. 1,

by its matrix continued-fractional expansion

$$K = D_{(0)} + K_{(1)} \frac{1}{D_{(0)} - K_{(1)}} D_{(0)}$$

$$= D_{(0)} \left[D_{(0)} - A_{(1)} - D_{(1)} \frac{1}{D_{(1)} - A_{(2)} - \dots} D_{(1)} \right]^{-1} D_{(0)}. \quad (15)$$

This completes our construction; we should only notice that the matrix continued-fractional quantities represent also the coefficients in the generalized Born series (10),²

$$\phi = u + u_{(1)} \frac{1}{D_{(0)} - K_{(1)}} D_{(0)}$$

$$+ u_{(2)} \frac{1}{D_{(1)} - K_{(2)}} D_{(1)} \frac{1}{D_{(0)} - K_{(1)}} D_{(0)} + \dots \quad (16)$$

This makes our method elegant and simple. In the present notation, also the difference between the scalar and matrix MCF formulas disappears.

ACKNOWLEDGMENTS

I would like to thank T. Sasakawa and especially J. Horáček for the inspiration and thorough discussions on their method.

¹J. Horáček and T. Sasakawa, Phys. Rev. A **28**, 2151 (1983).

²M. Znojil, J. Math. Phys. **18**, 717 (1977).